Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/22/2004
PI Name:	Garoff, Stephen Ph.D.		
Project Title:	Wetting Dynamics of Complex Fluids and Surfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	15213-3890	**Congressional District:**	14
Comments:	T		
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Jacqmin, David	**Contact Phone:**	216-433-5853
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Flight Program:			
Flight Assignment:	Note: Original end date was 11/27/2007.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Walker, Lynn (Carnegie Mellon University) Rame, Enrique (National Center for Microgravity Research)		
Grant/Contract No.:	NNC04GA34G		
Performance Goal No.:	2.7		
Performance Goal Text:			
Task Description:	Using very accurate optical imaging techniques and particle image velocimetry, we characterize the shape of the fluid/fluid interface and the flow fields in a microscopic region near moving contact lines. From these images, we test models of the hydrodyna mics in regions near the contact line where viscous forces dominate the fluid flow and in regions even closer to the contact line where a unique hydrodynamic regime must govern the fluid motion. By making systematic changes in the materials system under examination, we seek the physical origins of these unique hydrodynamics. Low gravity studies are essential to this program; bouyancy flows that would otherwise mask the effects of capillary and wetting forces are eliminated. Objective: The study will characterize the unique hydrodynamics occurring in a microscopic region near moving contact		

	lines and that controls macroscopic spreading of fluids on solid surfaces. Significance: The task significance is to (1) establish the correct method for measuring dynamic contact angle boundary condition, (2) provide a proper basis for predictive models of spreading, (3) test models of the hydrodynamics near, but not at, moving contact lines where viscous forces are dominant, and (4) identify the molecular-scale mechanisms governing the unique hydrodynamic regimes at moving contact lines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions adn elastic fluids; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating or printing, which employ dynamic wetting by surfactant solutions and non-Newtonian fluds.
Task Progress:	During the past year, we have investigated dynamic wetting by two classes of complex fluids: surfactant solutions and polymer melts. We have found complex behavior when a solid surface is pulled from a bath of surfactant solution. At some critical velocity, a film of fluid is entrained onto the solid surface as if emerges from the bath. We have found an abrupt transition in the critical velocity at a characteristic surfactant concentration. This transition appears in a wide range of systems. Below that film pulling velocity, we find the contact lines of surfactant solutions do not remain smooth nor retreat by steady motion across the surface. Rather, the contact line is mostly flat and moving steadily but micron scale fluctuations occur. These fluctuations suggest that, sporadically and locally, the surfactant molecules cannot restructure fast enough as they pass through the contact line (the edge of the fluid on the surface) to form an equilibrium structure. Our research not only elucidates the fundamental hydrodynamics which control wetting by surfactant solutions; it also provides the understanding needed to optimize and control the many technological processes, such as cleaning and coating, which employ dynamic wetting by surfactant solutions. Non-Newtonian fluids are employed in many technologies, yet their wetting behavior has received far less attention than Newtonian fluids. We have found that two room temperature polymer melts, polyisobutylene and polystyrene, do not exhibit the same dynamic wetting behavior as another polymer melt, polydimethylsiloxane. All three fluids appear Newtonian in standard rotational rheological measurements, but only the latter shows wetting behavior explained by a model which assumes Newtonian behavior of the fluid. The polyisobutylene and polystyrene show behavior that is indicative of weak elasticity in the fluid. These experiments show that dynamic wetting is more sensitive to this elasticity than standard rheometric techniques. Probing fluids with strong elasticity, we have found that a series of polymer mixtures, Boger fluids, deviate strongly from Newtonian wetting behavior and may exhibit a fluid instability near the moving contact line. This research has direct application in developing well-founded avenues to design of the many technologies (e.g. painting, ink jet printing) that employ non-Newtonian fluids.
Bibliography Type:	Description: (Last Updated: 07/02/2007)
Articles in Peer-reviewed Journals	X. Chen, E. Rame, S. Garoff "The Effects of Thin and Ultra-thin Liquid Films on Dynamic Wetting" Physics of Fluids, 16,287 , Jan-2004
Articles in Peer-reviewed Journals	E. Rame, S. Garoff, K.R. Willson "Characterizing the Microscopic Physics Near Moving Contact Lines Using Dynamic Contact Angle Data" Accepted for publication in Phys. Rev. E , Dec-2004
Articles in Peer-reviewed Journals	G.K. Seevaratnam, L. Walker, E. Rame, S. Garoff "Wetting by Simple Room Temerature Polymer Melts: Deviations from Newtonian Behavior" submitted for publication in J. Coll. Interf. Sci. , Dec-2004
Presentation	G. Seevaratnam, L. Walker, S. Garoff "Spreading of Model non-Newtonian Fluids" 75th Annual Meeting of the Society of Rheology Oct-2003
Presentation	S. Garoff "Physics of Dynamic Wetting" Lincoln Labs, Waltham MA Nov-2003
Presentation	E. Rame "Modeling Liquid Spreading: A Fluid Mechanical Approach" Dept. Chemical Engineering, Clarkson University Dec-2003
Presentation	S. Garoff "Physics of Dynamic Wetting" Cabot Corporation, Billerica, MA Dec-2003
Presentation	E. Rame "The Influence of Dynamic Contact Angle on wetting Dynamics" ALA Lab Automation, Boston Jun-2004
Presentation	G. Seevaratnam, L.M. Walker, S. Garoff and E. Rame "Wetting by Simple Room Temperature Polymer Melts: Deviations from Newtonian Behavior" 78th ACS Colloids and Surface Science Symposium, New Haven, CT Jun-2004
Presentation	K S. Varanasi, S. Garoff "Concentration Dependent Transition in the Critical Velocity of Film Pulling in Surfactant Solutions" 78th ACS, Colloid and Surface Science Symposium, New Haven, CT Jun-2004
Presentation	E. Rame "Modelado de Mojado de Sólidos por Líquidos: un enfoque Fluido-mecánico" Grupo de Medios Porosos (Porous Media Group), Physics Dept. of the School of Engineering, University of Buenos Aires, Buenos Aires Aug-2004